

MySQL Practice Exercises with Hints HR, Sales & Sample Databases

GUIDELINES TO PRACTICE

1. Read the question carefully and identify tables involved.
2. Draw simple table relationships (ERD) before writing queries.
3. Start with `SELECT *` to understand data.
4. Use JOIN conditions correctly (PK = FK).
5. Test queries step by step.
6. For triggers and procedures, always test with sample INSERT/UPDATE.
7. Use TRANSACTION when practicing triggers and deletes.
8. Comment your SQL for clarity.

SECTION 1 – JOINS (HR + SALES + SAMPLE DB)

1. HR: Display employee name and department name.
Hint: Join employees and departments using department_id.
2. HR: Find employees who don't belong to any department.
Hint: LEFT JOIN + WHERE dept_id IS NULL.
3. SALES: Show all customers and their orders.
Hint: LEFT JOIN customers → orders.
4. SALES: Find customers who never placed an order.
Hint: LEFT JOIN + filter NULL in orders.
5. SAMPLE: Show students with their enrolled courses.
Hint: Use INNER JOIN between students and courses.
6. HR: List employees with their manager names.
Hint: SELF JOIN employees table.
7. SALES: Display product name and total quantity sold.
Hint: Join products + order_details, use SUM.
8. HR: Find departments having more than 3 employees.
Hint: JOIN + GROUP BY + HAVING.
9. SALES: Show orders with customer name and payment mode.
Hint: Join orders, customers, payments.
10. SAMPLE: Show all possible combinations of students and clubs.
Hint: Use CROSS JOIN.

SECTION 2 – VIEWS

11. HR: Create a view showing only active employees.
Hint: WHERE status = 'Active'.
12. SALES: Create a view for monthly sales report.
Hint: Use GROUP BY MONTH(order_date).
13. HR: Create a view hiding salary column.
Hint: Select only non-sensitive fields.
14. SAMPLE: Create a view joining students and courses.
Hint: Base the view on JOIN query.
15. SALES: Create a read-only view of high value customers.
Hint: Use WITH CHECK OPTION.

SECTION 3 – TRIGGERS

16. HR: Log employee insert into audit table.
Hint: AFTER INSERT trigger, use NEW.column.
17. HR: Prevent deleting employees from Admin department.
Hint: BEFORE DELETE + SIGNAL error.
18. SALES: Reduce stock after order insert.
Hint: AFTER INSERT on order_details.

19. SALES: Prevent negative stock.
Hint: BEFORE UPDATE, check NEW.stock < 0.
20. SAMPLE: Store deleted student records in backup table.
Hint: BEFORE DELETE trigger, use OLD values.
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SECTION 4 – STORED PROCEDURES

21. HR: Procedure to show employees by department.
Hint: Use IN parameter for dept_id.
22. HR: Procedure to increase salary by percentage.
Hint: UPDATE with calculation.
23. SALES: Procedure to insert new order.
Hint: Use parameters for customer_id, date.
24. SALES: Procedure to calculate total sales between dates.
Hint: Use SUM with WHERE BETWEEN.
25. SAMPLE: Procedure to count students in a course.
Hint: Use COUNT() and GROUP BY.
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SECTION 5 – USER DEFINED FUNCTIONS

26. HR: Function to calculate age from DOB.
Hint: Use TIMESTAMPDIFF(YEAR, dob, CURDATE()).
27. SALES: Function to calculate GST.
Hint: Multiply amount * tax_rate.
28. SAMPLE: Function to return grade based on marks.
Hint: Use IF or CASE.
29. HR: Function to return full employee name.
Hint: CONCAT(first_name, ' ', last_name).
30. SALES: Function to check if customer is premium.
Hint: Use IF(total_purchase > 50000, 'YES', 'NO').